



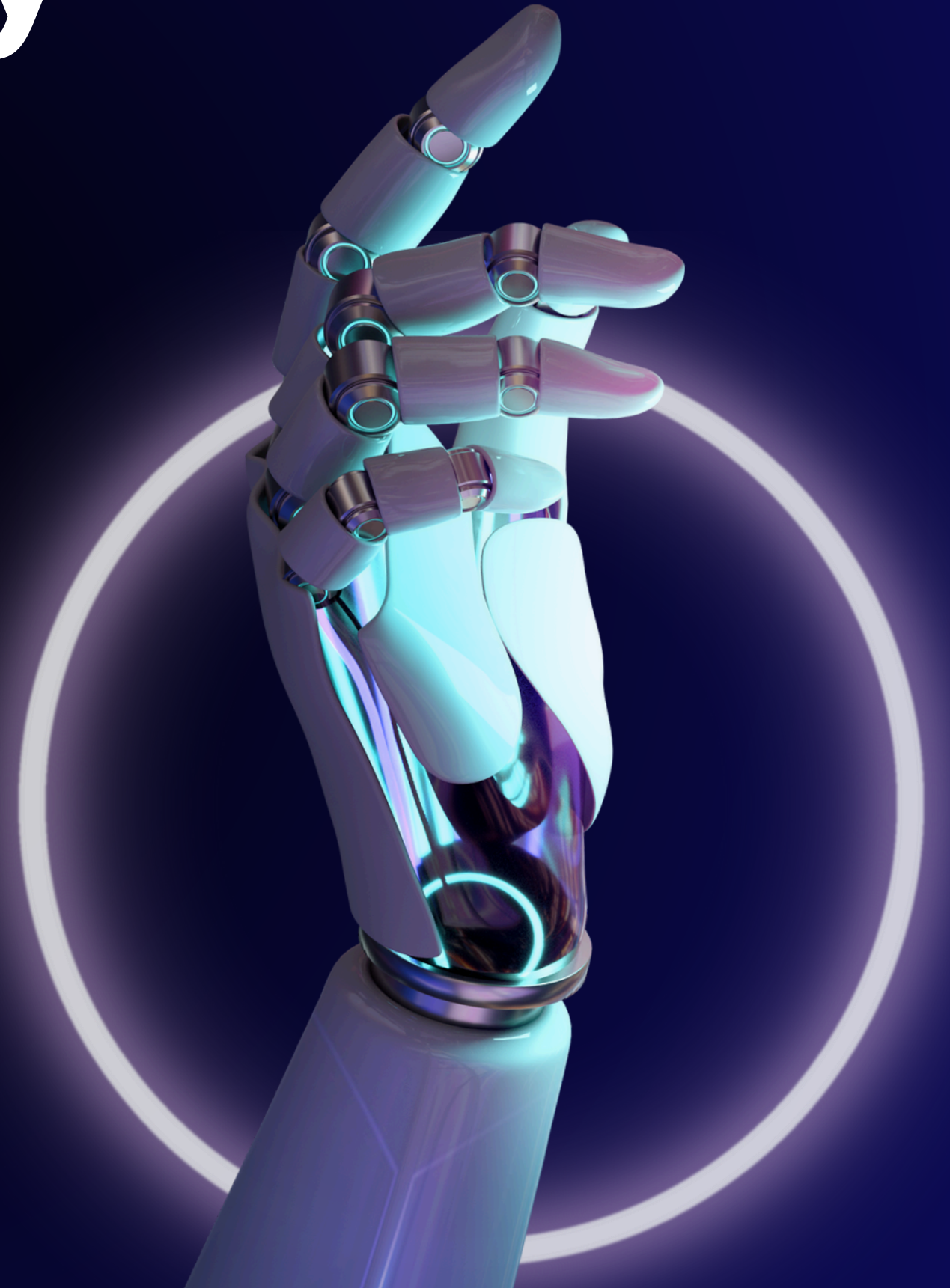
Sharkos Financials

This presentation will introduce our NLP Fine-tuning project, highlighting its objectives, methodologies, and potential impact.

Presented by Achille Krotoff, Félix Charotte, Jules Duval & Kewin Sungkur

Summary

1. LLMs in Finance Industry
2. Project introduction
3. Data Acquisition
4. Data Preparation
5. Fine-tuning





Project's goals

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Price fluctuations

Steps

- Data acquisition
- Preparation and integration
- Fine tuning and optimization

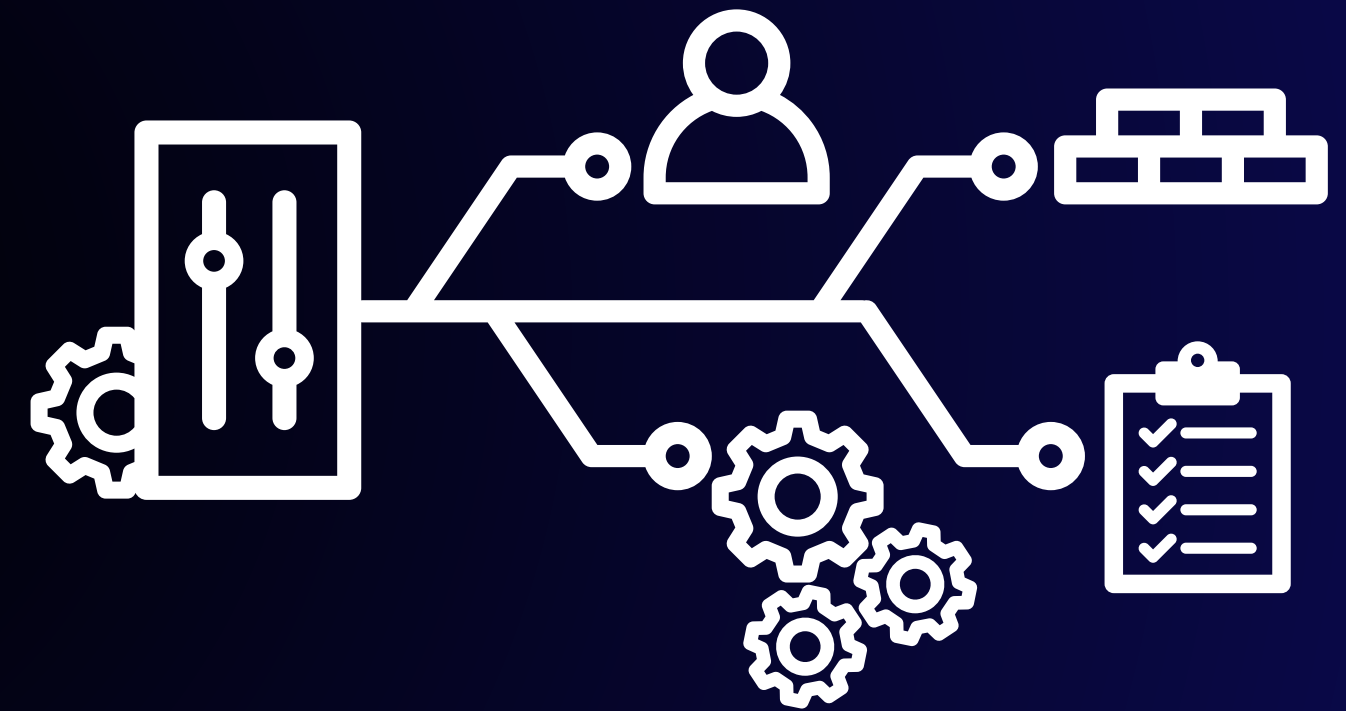


Data Acquisition

04

Data acquisition

- News (Scrapping -Hugging Face)
- Bitcoin price (API Binance)



Data selection

Data Preparation

1. Data Pre-Processing
2. Data merging
3. Categorization of price movements

Data Preparation

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Data Pre-processing

- News on Stock & Cryptocurrency Market

	_id	main_domain	title	description	created_at
0	6453d70d358e80adbfc4cb2b	cnbc.com	Dow drops 400 points, turns negative for the y...	Regional banks led the broader market lower as...	2023-05-04T16:01:46.448Z
1	6453cf909a78e3af538abe44	cointelegraph.com	Bitcoin drops with stocks as analyst warns of ...	Bitcoin dips as the U.S. banking crisis engulf...	2023-05-04T15:25:28.809Z
2	6453cb87ccab8508100df076	co.uk	Bitcoin Price Analysis: 29370 Tested After Su...	Bitcoin (BTC/USD) sought to add to recent gain...	2023-05-04T15:12:00.971Z
3	6453afd269f3c1643cf0a4f6	bitcoinist.com	Bitcoin Is 75% To Halving, Here's How Past Cyc...	The current Bitcoin cycle is now 75% on the wa...	2023-05-04T13:10:51.220Z
4	645399d92471d73ea0976d27	seekingalpha.com	Iron Mountain FFO of \$0.71 beats by \$0.03, rev...	Iron Mountain press release (IRM): Q1 FFO of...	2023-05-04T11:41:12.498Z

Filter, Date formatting,
Select columns...



	description	created_at
1	Bitcoin dips as the U.S. banking crisis engulf...	2023-05-04
2	Bitcoin (BTC/USD) sought to add to recent gain...	2023-05-04
3	The current Bitcoin cycle is now 75% on the wa...	2023-05-04
9	BTC/USD Stops Elected Below 28164: Sally Ho's...	2023-05-04
10	Trusted since 2012, BitKan cryptocurrency brok...	2023-05-04

- Bitcoin Price

Date	Price
2021-08-02 02:00:00	39147.82
2021-08-03 02:00:00	38207.05
2021-08-04 02:00:00	39723.18
2021-08-05 02:00:00	40862.46
2021-08-06 02:00:00	42836.87
2021-08-07 02:00:00	44572.54
2021-08-08 02:00:00	43794.37
2021-08-09 02:00:00	46253.4

Convert in percentage,
Select columns



Date	Price Variation
2021-08-02	1.01
2021-08-03	97.56
2021-08-04	1.04
2021-08-05	1.02
2021-08-06	1.04
2021-08-07	1.06
2021-08-08	97.56
2021-08-09	1.06

Data Preparation

Merging

- News on Stock & Cryptocurrency Market

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2021-08-09	1.06

Merged Dataset

Data Preparation

Categorization of price movements

- **Segment in 4 quartiles** : Extreme bearish, Bearish, Bullish, Extreme Bullish
- **Add Instructions Column**

1	input	output	Instruction
2	Bitcoin dips as the U.S. banking crisis engulfs more lenders, BTC price falling in line with stocks.	Bearish	Predict price variation in percentage of the Bictoin considering this recent market news
3	Bitcoin (BTC/USD) sought to add to recent gains early in the Asian session as the pair extended re	Bearish	Predict price variation in percentage of the Bictoin considering this recent market news
4	The current Bitcoin cycle is now 75% on the way to the next halving. Here's what previous cycles l	Bearish	Predict price variation in percentage of the Bictoin considering this recent market news
5	BTC/USD Stops Elected Below 28164: Sally Ho's Technical Analysis – 4 May 2023 Bitcoin (BTC/USC	Bearish	Predict price variation in percentage of the Bictoin considering this recent market news
6	Trusted since 2012, BitKan cryptocurrency broker exchange provides you a better way to buy, trac	Bearish	Predict price variation in percentage of the Bictoin considering this recent market news
7	Buyers will have to fiercely defend the respective support in Bitcoin and select altcoins if they war	Bullish	Predict price variation in percentage of the Bictoin considering this recent market news
8	Bitcoin avoids volatility into the weekly and monthly BTC price closes.	Bearish	Predict price variation in percentage of the Bictoin considering this recent market news
9	Cointelegraph analyst and writer Marcel Pechman explains how a U.S. debt default could impact l	Bearish	Predict price variation in percentage of the Bictoin considering this recent market news
10	The world was rocked by a sharp drop in the value of Bitcoin, which resulted in a Long Squeeze a	Extrem Bullish	Predict price variation in percentage of the Bictoin considering this recent market news

Fine tuning

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- Definition of the model and the tokenizer

```
from unsloth import FastLanguageModel
import torch
max_seq_length = 2048 # Choose any! We auto support RoPE Scaling internally!
dtype = None # None for auto detection. Float16 for Tesla T4, V100, Bfloat16 for Ampere+
load_in_4bit = True # Use 4bit quantization to reduce memory usage. Can be False.

model, tokenizer = FastLanguageModel.from_pretrained(
    model_name = "unsloth/llama-3-8b-Instruct-bnb-4bit",
    max_seq_length = max_seq_length,
    dtype = dtype,
    load_in_4bit = load_in_4bit,
    # token = "hf_...", # use one if using gated models like meta-llama/Llama-2-7b-hf
)
```

Model importation

Fine tuning

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- Add of the Lora Adaptaters

```
model = FastLanguageModel.get_peft_model(  
    model,  
    r = 16, # Choose any number > 0 ! Suggested 8, 16, 32, 64, 128  
    target_modules = ["q_proj", "k_proj", "v_proj", "o_proj",  
                     "gate_proj", "up_proj", "down_proj",],  
    lora_alpha = 16,  
    lora_dropout = 0, # Supports any, but = 0 is optimized  
    bias = "none",    # Supports any, but = "none" is optimized  
    # [NEW] "unsloth" uses 30% less VRAM, fits 2x larger batch sizes!  
    use_gradient_checkpointing = "unsloth", # True or "unsloth" for very long context  
    random_state = 3407,  
    use_rslora = False, # We support rank stabilized LoRA  
    loftq_config = None, # And LoftQ
```

)

Fine tuning

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- Pre processing of the data

```
alpaca_prompt = """Below is an instruction that describes a task, paired with an input that provides further context. Write a response that appropriately completes the request.

### Instruction:
{}

### Input:
{}

### Response:
{}"""

EOS_TOKEN = tokenizer.eos_token # Must add EOS_TOKEN
def formatting_prompts_func(examples):
    instructions = examples["Instruction"]
    inputs       = examples["input"]
    outputs      = examples["output"]
    texts = []
    for instruction, input, output in zip(instructions, inputs, outputs):
        # Must add EOS_TOKEN, otherwise your generation will go on forever!
        text = alpaca_prompt.format(instruction, input, output) + EOS_TOKEN
        texts.append(text)
    return { "text": texts, }
pass

from datasets import load_dataset
dataset = load_dataset("Achillektf/NLP_News_Cat", split = "train")
dataset = dataset.map(formatting_prompts_func, batched = True,)
```

Columns of the dataset

Fine tuning

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- Definition of the trainer

```
from trl import SFTTrainer
from transformers import TrainingArguments

trainer = SFTTrainer(
    model = model,
    tokenizer = tokenizer,
    train_dataset = dataset,
    dataset_text_field = "text",
    max_seq_length = max_seq_length,
    dataset_num_proc = 2,
    packing = False, # Can make training 5x faster for short sequences.
    args = TrainingArguments(
        per_device_train_batch_size = 2,
        gradient_accumulation_steps = 4,
        warmup_steps = 5,
        max_steps = 30,
        learning_rate = 2e-4,
        fp16 = not torch.cuda.is_bf16_supported(),
        bf16 = torch.cuda.is_bf16_supported(),
        logging_steps = 1,
        optim = "adamw_8bit",
        weight_decay = 0.01,
        lr_scheduler_type = "linear",
        seed = 3407,
        output_dir = "outputs",
    ),
)
```


Fine tuning

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- Test of the model :

```
# alpaca_prompt = Copied from above
FastLanguageModel.for_inference(model) # Enable native 2x faster inference
inputs = tokenizer(
    [
        alpaca_prompt.format(
            "Predict the price of the bictoin with the following news", # instruction
            "Blackrock buy 2 trillion of bitcoin",
            "", # output - leave this blank for generation!
        )
    ], return_tensors = "pt").to("cuda")

outputs = model.generate(**inputs, max_new_tokens = 64, use_cache = True)
tokenizer.batch_decode(outputs)
```

```
Setting `pad_token_id` to `eos_token_id`:128001 for open-end generation.
[ '<|begin_of_text|>Below is an instruction that describes a task, paired with an input that provides further context. Write a response that appropriately completes the request.\n\n### Instruction:\nPredict the price of the bictoin with the following news\n\n### Input:\nBlackrock buy 2 trillion of bitcoin\n\n### Response:\nExtrem Bullish<|eot_id|>']
```

Conclusion

1. Data Pre-Processing
2. Data merging
3. Categorization of price movements

Thank You!

Time for Q&A !

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